Machine Learning Project Ideas for Final Year and Research Publication

1. Explainable Fraud Detection using Machine Learning

This project aims to detect fraudulent financial transactions using classical and ensemble-based machine learning models. Special emphasis is placed on handling class imbalance and providing interpretable explanations using feature attribution techniques. The system improves transparency and trust, making it suitable for real-world financial applications.

2. Predictive Maintenance using Sensor Data Analytics

The objective is to predict machine failures by analyzing historical sensor data collected from industrial equipment. Time-series features and machine learning models are used to identify early signs of failure. The project focuses on practical deployment aspects and performance evaluation.

3. Automated Resume Screening using NLP

This project develops an automated system to match resumes with job descriptions using natural language processing techniques. Semantic similarity models are employed to rank candidates, reducing manual screening effort while ensuring consistency and efficiency in recruitment processes.

4. Medical Diagnosis Support System using Explainable ML

The goal is to build a medical diagnosis support tool using patient data or medical images. The system combines accurate prediction with explainable outputs, helping practitioners understand the reasoning behind model decisions and increasing clinical trust.

5. Traffic Congestion Prediction using Machine Learning

This project focuses on predicting traffic congestion levels using historical traffic data and environmental features. Machine learning models are used to forecast congestion patterns, supporting intelligent traffic management and urban planning systems.

6. Fake News Detection using Text-Based Machine Learning

The objective is to identify fake news articles using textual features and supervised learning techniques. The project evaluates different feature representations and classifiers to improve robustness and generalization across datasets.

7. Customer Churn Prediction with Business Insights

This project predicts customer churn using historical usage and behavioral data. Feature importance analysis is conducted to identify key churn factors, enabling actionable insights for improving customer retention strategies.

8. Recommendation System using Collaborative Filtering

The aim is to develop a recommendation system that suggests products or content based on user preferences. Collaborative filtering techniques are implemented and evaluated, with a focus on scalability and performance.

9. Network Intrusion Detection using Machine Learning

This project builds an intrusion detection system by analyzing network traffic data to identify malicious activities. Machine learning classifiers are trained to distinguish normal and abnormal behavior, improving cybersecurity monitoring.

10. Document Classification and Information Extraction System

The objective is to automatically classify documents and extract key information using machine learning and NLP techniques. The system reduces manual document processing effort and improves efficiency in enterprise workflows.

These projects are designed to balance implementation feasibility with research novelty, making them suitable for final-year evaluation and academic publication.